Sri Chaitanya IIT Academy., India.

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A right Choice for the Real Aspirant

Central Office, Bangalore

Sec: (Inc)Sr. Super60 (S) & C120 — Date: 14-03-2026

Time: 09.00 A.M TO 12.00 P.M — Max. Marks: 300

JEE-MAIN

KEY SHEET

MATHEMATICS

1	**1**	2	**4**	3	**3**	4	**1**	5	**2**
6	**2**	7	**1**	8	**1**	9	**1**	10	**2**
11	**1**	12	**2**	13	**1**	14	**1**	15	**2**
16	**4**	17	**1**	18	**1**	19	**2**	20	**3**
21	**10**	22	**0**	23	**9**	24	**8**	25	**8s**

PHYSICS

26)	**3**	27	**3**	28	**2**	29	**3**	30	**1**
31	**4**	32	**1**	33	**1**	34	**1**	35	**2**
36	**3**	37	**1**	38	**2**	39	**1**	40	**2**
41	**3**	42	**2**	43	**1**	44	**1**	45	**2**
46	**11**	47	**1**	48	**2**	49	**65**	50	**25**

CHEMISTRY

51)	**3**	52	**4**	53	**4**	54	**3**	55	**2**
56	**4**	57	**3**	58	**3**	59	**4**	60	**2**
61	**2**	62	**1**	63	**1**	64	**4**	65	**3**
66	**1**	67	**2**	68	**1**	69	**2**	70	**3**
71	**2**	72	**7**	73	**7**	74	**4**	75	**4**

SOLUTION
MATHEMATICS

1. $\int \frac{\sin^2 x - \cos^2 x}{\sin^2 x \cos^2 x} dx$

$\int \frac{\sin^2 x}{\sin^2 x \cos^2 x} - \frac{\cos^2 x}{\sin^2 x \cos^2 x} dx$

$\Rightarrow \int \left(\sec^2 x - \cos e^2 x\right)$

$\Rightarrow \int \sec^2 x\, dx - \int \cos ec^2 x\, dx$

$\Rightarrow \tan x - (-\cot x) + C$

$\Rightarrow \tan x + \cot x + C$

2. $\int \frac{10x^9 + 10x \log e^{10}}{x^{10} + 10x} dx$

$let\ f(x) = x^{10} + 10^x$ $\qquad \int \frac{f'(x)}{f(x)} dx = \log|f(x)| + C$

$f'(x) = 10x^9 + 10^x \log e^{10}$

$\log\left|x^{10} + 10^x\right| + C$

3. $I = \int \frac{dx}{(x+2)(x^2+1)} dx$

$\frac{1}{(x+2)(x^2+1)} = \frac{A}{x+2} + \frac{Bx+C}{x^2+1}$

Solving We get

$A = \frac{1}{5} \quad B = \frac{-1}{5} \quad C = \frac{2}{5}$

$\Rightarrow \frac{1}{5}\int \frac{1}{x+2} - \frac{1}{10}\int \frac{2x}{1+x^2} dx + \frac{1}{5}\int \frac{2}{1+x^2} dx$

$\Rightarrow \frac{1}{5}\log|x+2| - \frac{1}{10}\log\left|1+x^2\right| + \frac{2}{5}\log\left|1+x^2\right| + C$

$a = \frac{-1}{10}$ $\qquad$ $b = \frac{2}{5}$

4. $\int \frac{\cos 2x - \cos 2\theta}{\cos x - \cos\theta} dx$

$\Rightarrow 2\int \frac{\cos^2 x - \cos^2\theta}{\cos x - \cos\theta} dx$

$\Rightarrow 2\int (\cos x + \cos\theta) dx$

$\Rightarrow 2(\sin x + x\cos\theta) + C$

5. $\int (\sin x) / (3 + 4\cos^2 x)\, dx$

Let t = cos x

Then dt = -sin x dx

$\int (\sin x)/(3 + 4\cos^2 x)\, dx$

$= -\int dt / (3 + 4t^2)$

Here:

$3 + 4t^2 = (\sqrt{3})^2 + (2t)^2$

So,

$\int dt/(3 + 4t^2) = 1/(2\sqrt{3}) \tan^{-1}(2t/\sqrt{3})$

$= -1/(2\sqrt{3}) \tan^{-1}(2t/\sqrt{3}) + C$

$-1/(2\sqrt{3}) \tan^{-1}(2\cos x/\sqrt{3}) + C$

6. $\int \frac{x}{(x-1)(x-2)} dx$

$\frac{x}{(x-1)(x-2)} = \frac{A}{(x-1)} + \frac{B}{(x-2)}$

$A = -1$

$B = 2$

$\Rightarrow -\log(x-1) + 2\log(x-2) + C$

$$\Rightarrow \log(x-2)^2 - \log(x-1) + C$$

$$\log\left|\frac{(x-2)^2}{x-1}\right| + C$$

7. $\sin^3 x = \sin x\,(1 - \cos^2 x)$

Let $u = \cos x$

Then $du = -\sin x\, dx$

So, $\sin x\, dx = -du$

$\int \sin x\,(1 - \cos^2 x)\cos^2 x\, dx$

$= -\int (1 - u^2)u^2\, du$

$= -\int (u^2 - u^4)\, du$

$= -(u^3/3 - u^5/5) + C$

$= -u^3/3 + u^5/5 + C$

$= -(1/3)\cos^3 x + (1/5)\cos^5 x + C$

8. $\int \frac{dx}{e^x + e^{-x}} dx$

$$\Rightarrow \int \frac{e^x}{e^x.e^x + e^x e^{-x}} dx = \int \frac{e^x}{e^{2x}+1} dx$$

$Let\ u = e^x$

$$\Rightarrow \int \frac{u}{u^2+1} du = \tan^{-1}(u) + C$$

$$= \tan^{-1}(e^x) + C$$

9. $x^2 / (1 - x^4) = A/(1 - x) + B/(1 + x) + (Cx + D)/(1 + x^2)$

After solving, we get:

$= 1/4(1/(1 - x) + 1/(1 + x)) - 1/2\,(x/(1 + x^2))$

$\int dx/(1 - x) = -\log|1 - x|$

$\int dx/(1 + x) = \log|1 + x|$

$\int x/(1 + x^2)\, dx = 1/2 \log(1 + x^2)$

$= 1/4 \log |(1 + x)/(1 - x)| - 1/2 \tan^{-1} x + C$

10. $\int\left(\frac{2x+3-4}{2x+3}\right)dx$

$=\int 1\,dx-4\int\frac{1}{2x+3}dx$

$=x-\log\left|(2x+3)^2\right|+C$

11. $\int\frac{1}{1+\cos x}dx=\int\frac{1}{2\cos^2\frac{x}{2}}dx=\frac{1}{2}\int\sec^2\frac{x}{2}dx=\tan\frac{x}{2}+C$

12.

$\int\frac{\cos 2x}{(\sin x+\cos x)^2}dx=\int\frac{\cos^2 x-\sin^2 x}{(\sin x+\cos x)^2}dx=\int\frac{(\cos x+\sin x)(\cos x-\sin x)}{(\sin x+\cos x)^2}dx=\log|\sin x+\cos x|+C$

13. $\int\sqrt{1+x^2}\,dx$

$\int\sqrt{1+x^2}\,dx=\frac{x}{2}\sqrt{1+x^2}+\frac{1}{2}\log\left|x+\sqrt{1+x^2}\right|+C$

14. $I=\int\frac{1}{x^2(x^2+1)}dx \qquad x^2=t$

$I=\frac{1}{2}\int\frac{1}{t(t+1)}dt \qquad \Rightarrow \frac{1}{2}\int\frac{1}{t}dt-\frac{1}{2}\int\frac{1}{t+1}dt$

$=\frac{1}{2}\log t-\frac{1}{2}\log|t+1|+C$

$I=\frac{1}{2}\log x^2-\frac{1}{2}\log|x^2+1|+C$

$I=\log x-\frac{1}{2}\log|x^2+1|+C$

15. $I=\int \frac{dx}{x^2+2x+2}$

$$I=\int \frac{dx}{(x+1)^2+1^2}$$

$$I=\tan^{-1}(x+1)+C$$

16. $I=\int \frac{x+\sin x}{1+\cos x}dx$

$$I=\int \frac{x}{2\cos^2\frac{x}{2}}dx+\frac{2\sin\frac{x}{2}\cos\frac{x}{2}}{2\cos^2\frac{x}{2}}dx$$

$$=\frac{1}{2}\int x\sec^2\frac{x}{2}dx+\int \tan\frac{x}{2}dx$$

$$=\frac{1}{2}\left[x\,2\tan\frac{x}{2}-2\int \tan\frac{x}{2}dx\right]+\int \tan\frac{x}{2}dx$$

$$=x\tan\frac{x}{2}-\int \tan\frac{x}{2}dx+\int \tan\frac{x}{2}dx+C$$

$$I=x\tan\frac{x}{2}+C$$

17. $\int \frac{x+3}{(x+4)}e^x\,dx$

$$I=\int \frac{x+3}{(x+4)^2}e^x\,dx$$

$$=\int \frac{x+4-1}{(x+4)^2}e^x\,dx$$

$$=\int\left(\frac{1}{x+4}-\frac{1}{(x+4)^2}\right)e^x\,dx=e^x\left(\frac{1}{x+4}\right)+C$$

18. $Put\cos^2(2x)=t\,so\,that\,4\sin 2x\,dx=-dt$

$$I=\frac{1}{4}\int\frac{dt}{\sqrt{9-t^2}}=-\frac{1}{4}\sin^{-1}\left(\frac{1}{3}\right)+C=-\frac{1}{4}\sin^{-1}\left[\frac{1}{3}\cos^2 2x\right]+C$$

19. $I=\int\left(\frac{1+\cos x}{x+\sin x}\right)dx$

$x+\sin x=t\Rightarrow 1+\cos x dx=dt$

$I=\int\frac{1}{t}dt=\log|t|+C$

$\log|x+\sin x|+C$

20. $y=\int\frac{\sin x+\cos x}{\sqrt{\cos^2 x+\sin^2 x+2\sin x\cos x}}dx$

$y=\int\frac{\sin x+\cos x}{\sin x+\cos x}dx$

$=x\,dx=x+C$

21. $\frac{x^2+1}{x^2-5x+6}=1+\left(\frac{5x-5}{x^2-5x+6}\right)$.....(1)

Also

$\frac{5x-5}{x^2-5x+6}=\frac{5x-5}{(x-2)(x-3)}$

let $\frac{5x-5}{x^2-5x+6}=\frac{A}{x-2}+\frac{B}{x-3}$

$5x-5=A(x-3)+B(x-3)$

let $x=3$ *let* $x=2$

$10=B$ $A=-5$

$\frac{5x-5}{x^2-5x+6}=\frac{-5}{x-2}+\frac{10}{x-3}$.....(2)

from 1 ,2

$I=\int dx-5\int\frac{dx}{x-2}+10\int\frac{dx}{x-3}$

$$= x - 5\log|x-2| + 10\log|x-3| + C$$

22. $I = \int \frac{x^3}{\sqrt{1+x^2}}dx = \int \frac{x^2.x}{\sqrt{1+x^2}}dx$

$$1 + x^2 = t^2$$

$$2x\,dx = 2t\,dt$$

$$I = \int \frac{t\left(t^2-1\right)}{t}dt = \frac{t^3}{3} - t + C$$

$$= \frac{1}{3}\left(1+x^2\right)^{\frac{3}{2}} - \sqrt{1+x^2} + C$$

$$a = \frac{1}{3} \text{ and } b = -1$$

23. $\int \frac{x^6 - x^5}{x^4 - x^3}dx$

$$\int x^2 dx + C$$

$$\frac{x^3}{3} + C$$

24. $I = \int \tan^2 x.\sec^4 x\,dx$

$$= \int \tan^2 x.\sec^2 x\,dx$$

$$= \int \tan^2 x\left(1+\tan^2 x\right)\left(\sec^2 x\right)dx$$

$$\Rightarrow \tan x = t$$

$$\sec^2 x\,dx = dt$$

$$I = \int t^2\left(1+t^2\right)dt$$

$$I = \int t^2 + t^4\,dt$$

$$I = \frac{t^3}{3} + \frac{t^5}{5} + C$$

$$I=\frac{\tan^5 x}{5}+\frac{\tan^3 x}{3}+C$$

25. $\int \frac{1}{\sqrt{9x-4x^2}}dx$

$$\Rightarrow \frac{1}{2}\int \frac{1}{\sqrt{\left(\frac{9}{8}\right)^2-\left[x^2-\frac{9}{4}x+\left(\frac{9}{8}\right)^2\right]}}dx$$

$$\Rightarrow \frac{1}{2}\int \frac{1}{\sqrt{\left(\frac{9}{8}\right)^2-\left(x-\frac{9}{8}\right)^2}}dx$$

$$\Rightarrow \frac{1}{2}\sin^{-1}\left(\frac{8x-9}{9}\right)+C$$

PHYSICS

26. The magnetic field due to straight segment is zero

27. $r=\frac{p}{Bq}=\frac{\sqrt{2mK}}{Bq}\Rightarrow r\ \alpha\ \frac{\sqrt{m}}{q}$

28. Since the coil is tightly wound, we may take each circular element to have the same radius R= 10 cm =0.1 m. The number of turns N=100 . The magnitude of the magnetic field is

$$B=\frac{\mu_0 NI}{2R}=\frac{4\pi\times10^{-7}\times10^2\times1}{2\times10^{-1}}=2\pi\times10^{-4}=6.28\times10^{-4}T$$

29. Using equation we find

$$r=\frac{mv}{(qB)}=\frac{9\times10^{-31}kg\times3\times10^7 ms^{-1}}{\left(1.6\times10^{-19}C\times6\times10^{-4}T\right)}=28\times10^{-2}m=28cm$$

$$v=\frac{v}{(2\pi r)}=17\times10^6 s^{-1}=17\times10^6 Hz=17\ MHz$$

30. Conceptual

31. $M=NiA=Ni\pi r^2\Rightarrow M\ \alpha r^2$

32. Conceptual

33. $i_2=\frac{v}{a}=\frac{5}{300}=\frac{1}{60}amp$

$$s=\frac{G}{\left(\frac{1}{t_g}-1\right)}=\frac{300}{\frac{5}{\frac{1}{60}}-1}=1\Omega$$

Now $S120\ A\times10^{-4}=1\times120\times5\times10^{-4}=0.06$

34. The field at the centre of circular loop is $B=\frac{\mu i}{2R}$ and the currents are alternately in opposite direction

35. The magnetic field at any point on the closed loop is due to all the three currents But line integral of i_3 over the closed loop will be zero

36. $\frac{\phi}{I}=\frac{BAN}{c}\Rightarrow\frac{\phi}{V}=\frac{BAN}{c}\frac{I}{V}=\frac{BAN}{c}\times\frac{1}{R}$

If current sensitivity doubled then resistance also doubled, So voltage sensitivity does not change

37. Due to flow of current in same direction at adjacent side an attractive magnetic force Will be produced.

38. Case a: $B_A=\frac{\mu_0}{4\pi}\frac{I}{l}\times(2\pi)^2$

Case B: $B_B=4\times\frac{\mu_0}{4\pi}\frac{1}{a/2}\left[\sin45^{\circ}+\sin45^{0}\right]$

39. Current carrying outwards to the plane we know that direction of B is perpendicular to both current and position vector so B at centre is zero

40. $B=\frac{\mu_0 i_1 i_2}{2\pi}\left[\frac{1}{r_1}-\frac{1}{r_2}\right]$

$=2\times10^{-7}\times20\times10\left[\frac{1}{1}-\frac{1}{10}\right]\times\frac{1}{10^{-2}}\times20\times10^{-2}$

$=7.2\times10^{-4}N$ (attraction)

41. For range 1 volt, $I_g=\frac{V}{(G+R_1)}$

$\therefore10^{-3}=\frac{1}{(50+R_1)}\ or\ 50+R_1=1000$

$R_1=1000-50=950\Omega$

42. $B=\left(\frac{\mu_0 I}{2R^2}\right)r\quad(r<R)\Rightarrow B\ \alpha\ r$

$and\ B=\frac{\mu_0 I}{2\pi r}(r>R)\Rightarrow B\alpha\frac{1}{r}$

43. $a)F=Bi(2r)=(1.5)(2)(2\times0.5)=3N$

$b)F=Bi(\lambda)=(1.5)(2)(2)=6N$

$c)F=Bi\left(2l\sin\frac{\theta}{2}\right)=(1.5)(2)\left(2\times2\times\frac{1}{2}\right)=6N$

$d)F=Bi(l_e)=(1.5)(2)(0)=0N$

44. Conceptual

45. For long solenoid at the centre $B=\mu_0 ni$

$$H = \frac{B}{\mu_0} = ni = (500\times 2)5\times 5 = 2.5\times 10^4 \frac{A}{m}$$

46. $x_1 = R\frac{l_1}{l_2}$

$$x_1 = 60\times\frac{60}{40} = 90\Omega$$

$$x_2 = R\frac{l_1^{'}}{l_2^{'}} = 60\times\frac{40}{60} = 40s$$

$$\frac{x_1}{x_2} = \frac{90}{40} = 2.25 \; then \; 4n = 11$$

47. $i = \frac{E_1 - E_2}{r_1 + r_2}$

$$= \frac{20-10}{4+6} = 1A$$

48. $G = 98\Omega; \frac{i_g}{i}\times 100 = 2\%$

$$s = \frac{G}{\left(\frac{i}{i_g} - 1\right)} \therefore \frac{i}{i_g} = \frac{100}{2} = 50 \quad \therefore S = \frac{98}{(50-1)} = 2\Omega$$

49. We find that there is an upward force F, of magnitude IlB For mid air suspension, this must be balanced by the force due to gravity

$$mg = IlB$$

$$B = \frac{mg}{Il}$$

$$= \frac{0.2\times 9.8}{2\times 1.5} = 0.65T$$

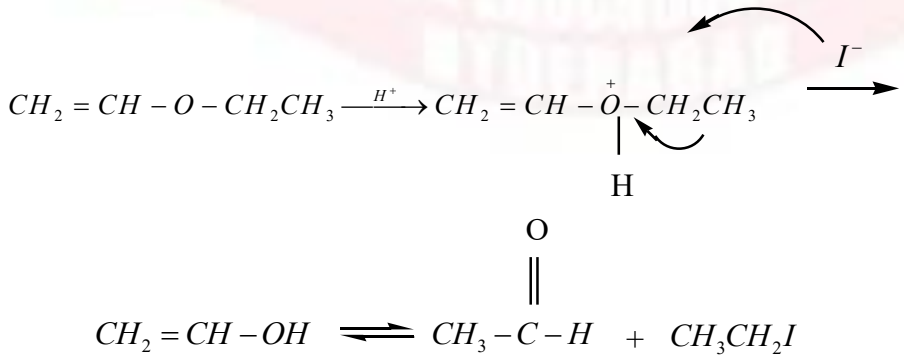
50. $\frac{1}{2}mv^2 = \vec{F}.\vec{s}; \frac{1}{2}m(25) = q(\vec{E}.\vec{S})$

$$\frac{25}{2} = \alpha(e\hat{i}.x\hat{i}); ex\alpha = \frac{25}{2} \quad x = \frac{25}{2\alpha e}$$

CHEMISTRY

51, Ethanol is dehydrated to ethene in the presence of H_2SO_4 at 443K. At $413K$ Ethoxyethane is main product

52.



53. Conceptual
54.

$$Ph-CH_3+CrO_2Cl_2 \rightarrow Ph-CH\left(OCrOHCl_2\right) \xrightarrow{H_3O^+} Ph-CHO$$

Chromium complex

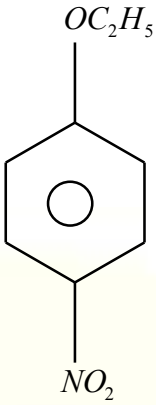
55. Hydrogenbond with water
56. Conceptual
57. X-yields phenylhydrazone and –ve response to the iodoform test and Tollen's test

It must contain a $>C=O$ group but is neither a methylketone nor aldehyde.

X could be $CH_3-CH_2-\overset{\overset{\displaystyle O}{\|}}{C}-CH_2-CH_3$

58. Hydrogen bonding in alcohols
59. A- 4, B - 5, C - 2, D - 1
60. $RCN+SnCl_2+HCl \rightarrow RCH=NH \xrightarrow{H_3O^+} RCHO$
61. aldehydes are more reactive than ketones in nucleophilic addition reactions due To storic and electronic reasons
62. Hydrogen sulphite adduct is water soluble and gives back original compounds by Treating with mineral acid or alkali
63. -OH group strongly activating group than $-OCH_3$ group
64. EWG increases the rate of reaction
65. BP : $R-OH > R-\overset{\overset{\displaystyle O}{\|}}{C}-H > R-O-R > R-H$
66. $2I^- \rightarrow I_2+2e^-$

    $etherperoxide+2e^- \rightarrow ether+O_2$
67. $Ph-OC_2H_5+HBr \xrightarrow{373K} Ph-OH+CH_3CH_2Br$

$Ph-OC_2H_5 + Con.H_2SO_4 + Conc:HNO_3 \rightarrow$ 

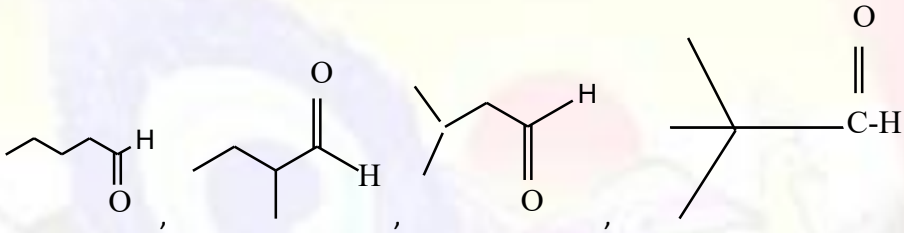
(major)

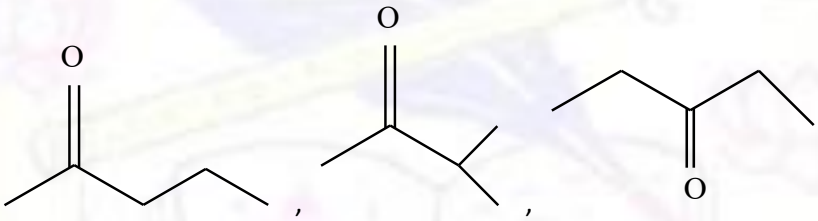
68. 2^0 & 3^0 alcohols give alkenes by elimination

69. Presence of strong base undergoes dehydrohalogenation

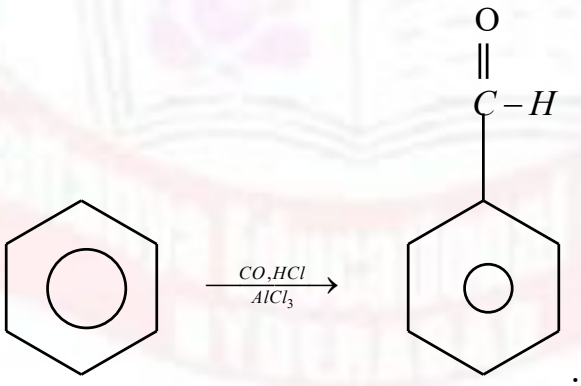
70. Popoff's rule

71. $(C_6H_5)_2O$ and $(CH_3)_3COC(CH_3)_3$

72. Aldehydes: 

Ketones: 

73

$HC \equiv CH \xrightarrow[\text{Fe tube}]{\text{Red hot}}$ 

74. All reactions

75 i, iii, iv, v

s